PROJECT PORTFOLIO

User Retention Analysis

Temporal Cohort Modeling & Heatmap Visualization using Python

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Business Intelligence at MSME Laundry Services

SWIPE <<<

INTRODUCTION

Project Overview

INDUSTRY:
MSME Laundry Services

Fast-growing micro, small, and medium-sized enterprise (MSME) industry in Indonesia, driven by urban mobility, busy lifestyles, and the need to outsource daily household chores.

ROLE:
Business Intelligence

Turn daily operational data from September 2025 to February 2026, spanning late-year activity (Sep–Dec 2025) and early-year trends (Jan–Feb 2026), into actionable strategies that increase customer lifetime value, optimize machine usage, and protect margins.

OBJECTIVE:
Implement Cohort Analysis

The core revenue driver for laundry services is repeat business. Acquiring a new customer is expensive, but keeping an existing customer coming back drives profitability.

CONCEPT DEVELOPMENT

Methodology

Raw Transaction Data

Extracted date and customer identifiers, parsed UTC timestamps, and standardized calendar features.

Cohort Formation

Identified each customer's initial transaction period and computed relative month index offsets.

Retention Calculation

Built a matrix of unique active users per cohort and calculated percentage retention relative to Month 0.

Visualization

Generated an annotated retention matrix heatmap using seaborn to isolate drop-off points.

Key Findings & Business Implications

Technical Implementation

Transaction Data

Loaded Excel transaction sheet into a pandas DataFrame, filtered out only the primary columns (Date and Customer ID).
Converted the text dates into standard UTC datetime objects. From those dates, extract separate year and month numbers.

```
[ ]
import pandas as pd

# 1. importing data
df = pd.read_excel(r'C:\Backup_D\Project\rekap_transaksi.xlsx', sheet_name=4)
df.head()

# 2. choose column that needed
df2 = df[['Date', 'Customer ID']].copy()
df2.head()

# 3. standardized calendar features
df2['datetime'] = pd.to_datetime(df['Date'], utc=True)
df2['month'] = df2['datetime'].dt.month
df2['year'] = df2['datetime'].dt.year

df2.head()
```

Python

```
# 4. create cohort
df2['cohort'] = (
    df2.apply(lambda row: (row['year'] * 100)
    +
    (row['month']), axis=1)
)

df2.head()

# 5. create first cohort
cohorts = (
    df2
    .groupby('Customer ID')['cohort']
    .min()
    .reset_index()
)
cohorts.columns = ['Customer ID', 'first_cohort']

df2 = df2.merge(
    cohorts,
    on='Customer ID',
    how='left'
)

df2.head()
```

[ ] Python

Cohort Formation

Create a numerical cohort tag for every transaction in the format *YYYYMM* (multiplying the year by 100 and adding the month). Then, groupe the data by customer Name and found the earliest cohort date for each person, labeling it as their *first_cohort*. Finally, merge this back into the main *DataFrame* so every transaction knew when that customer first started.

Retention Calculation

Calculated the time gap *cohort_distance* between a customer's transaction month and their very first visit month. Then, built a pivot table counting unique customers (*nunique*) across each cohort row and distance column, and divided each row by its Month 0 base (*cohort_pv[0]*) to turn the raw counts into clear retention percentages.

```
[ ]
# 6. create cohort retention
headers = sorted(df2['cohort'].dropna().unique().tolist())
df2['cohort_distance'] = df2.apply(
    lambda row: (headers.index(row['cohort']) - headers.index(row['first_cohort']))
    if pd.notna(row['first_cohort']) and pd.notna(row['cohort'])
    else None,
    axis=1
)

df2.tail()

# 7. built a pivot table
cohort_pv = pd.pivot_table(
    df2,
    index='first_cohort',
    columns='cohort_distance',
    values='Customer ID',
    aggfunc='nunique'
)

cohort_pv

# 8. divide each row
cohort_pv = cohort_pv.div(cohort_pv[0], axis=0)

cohort_pv
                                                                    Python
```

```
[ ]
```

```python
import seaborn as sns
import matplotlib.pyplot as plt

# 9. create visualization
fig_dims = (12,8)
fig, ax = plt.subplots(figsize=fig_dims)
sns.heatmap(
    cohort_pv,
    annot=True,
    fmt='.0%',
    mask=cohort_pv.isnull(),
    ax=ax,
    square=True,
    linewidths=.5,
    cmap=sns.cubehelix_palette(rot=-.2)
)

plt.show()
```

Python

Visualization

Normalized retention matrix into Seaborn and Matplotlib to plot a clean, color-coded heatmap with percentage annotations (*fmt='.0%'*), hiding empty future months using a mask.

OUTPUT SYNTAX

RESULT

	Date	Order ID	Customer ID	Weight	Piece	Piece Type	Service Speed	Service	Sales	Region
0	2025-09-01	1001	INE729	0.96	NaN	NaN	Reguler	Complete Wash	24.00	GJM
1	2025-09-01	1002	PER783	3.38	NaN	NaN	Reguler	Complete Wash	27.04	GJM
2	2025-09-01	1002	PER783	0.00	Small Bed Sheet Set	Bed Sheet	Piece	Piece	15.00	GJM
3	2025-09-01	1003	PUR183	4.49	NaN	NaN	Reguler	Complete Wash	35.92	GJM
4	2025-09-01	1003	PUR183	0.00	Small Bed Sheet Set	Bed Sheet	Piece	Piece	15.00	GJM

1. importing data

	Date	Customer ID
0	2025-09-01	INE729
1	2025-09-01	PER783
2	2025-09-01	PER783
3	2025-09-01	PUR183
4	2025-09-01	PUR183

2. choose primary column

	Date	Customer ID	datetime	month	year
0	2025-09-01	INE729	2025-09-01 00:00:00+00:00	9	2025
1	2025-09-01	PER783	2025-09-01 00:00:00+00:00	9	2025
2	2025-09-01	PER783	2025-09-01 00:00:00+00:00	9	2025
3	2025-09-01	PUR183	2025-09-01 00:00:00+00:00	9	2025
4	2025-09-01	PUR183	2025-09-01 00:00:00+00:00	9	2025

3. standardized calendar features

	Date	Customer ID	datetime	month	year	cohort
0	2025-09-01	INE729	2025-09-01 00:00:00+00:00	9	2025	202509
1	2025-09-01	PER783	2025-09-01 00:00:00+00:00	9	2025	202509
2	2025-09-01	PER783	2025-09-01 00:00:00+00:00	9	2025	202509
3	2025-09-01	PUR183	2025-09-01 00:00:00+00:00	9	2025	202509
4	2025-09-01	PUR183	2025-09-01 00:00:00+00:00	9	2025	202509

4. create cohort

	Date	Customer ID	datetime	month	year	cohort	first_cohort
0	2025-09-01	INE729	2025-09-01 00:00:00+00:00	9	2025	202509	202509
1	2025-09-01	PER783	2025-09-01 00:00:00+00:00	9	2025	202509	202509
2	2025-09-01	PER783	2025-09-01 00:00:00+00:00	9	2025	202509	202509
3	2025-09-01	PUR183	2025-09-01 00:00:00+00:00	9	2025	202509	202509
4	2025-09-01	PUR183	2025-09-01 00:00:00+00:00	9	2025	202509	202509

5. create first cohort

	Date	Customer ID	datetime	month	year	cohort	first_cohort	cohort_distance
9333	2026-02-28	JAV129	2026-02-28 00:00:00+00:00	2	2026	202602	202509	5
9334	2026-02-28	ANI982	2026-02-28 00:00:00+00:00	2	2026	202602	202509	5
9335	2026-02-28	DHA526	2026-02-28 00:00:00+00:00	2	2026	202602	202509	5
9336	2026-02-28	SHA685	2026-02-28 00:00:00+00:00	2	2026	202602	202509	5
9337	2026-02-28	RAH899	2026-02-28 00:00:00+00:00	2	2026	202602	202510	4

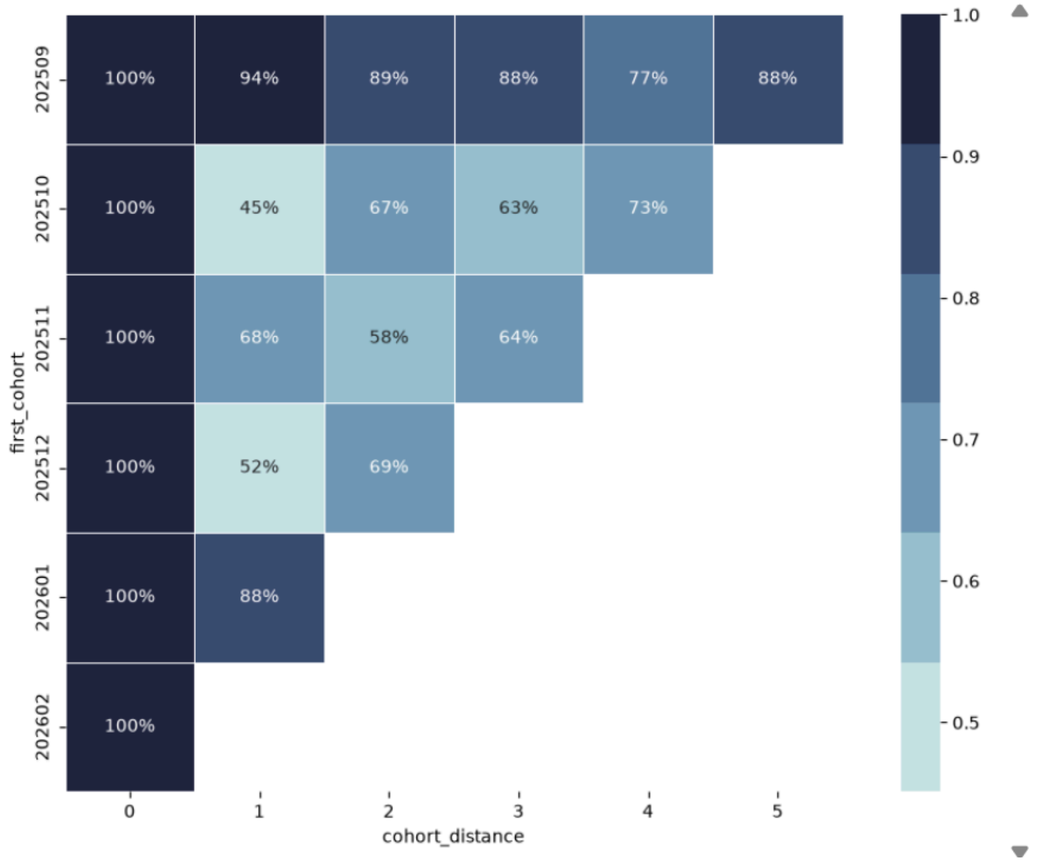
6. create cohort retention

cohort_distance	0	1	2	3	4	5
first_cohort						
202509	301.0	283.0	267.0	265.0	231.0	265.0
202510	51.0	23.0	34.0	32.0	37.0	NaN
202511	53.0	36.0	31.0	34.0	NaN	NaN
202512	29.0	15.0	20.0	NaN	NaN	NaN
202601	8.0	7.0	NaN	NaN	NaN	NaN
202602	7.0	NaN	NaN	NaN	NaN	NaN

7. built a pivot table

cohort_distance	0	1	2	3	4	5
first_cohort						
202509	1.0	0.940199	0.887043	0.880399	0.767442	0.880399
202510	1.0	0.450980	0.666667	0.627451	0.725490	NaN
202511	1.0	0.679245	0.584906	0.641509	NaN	NaN
202512	1.0	0.517241	0.689655	NaN	NaN	NaN
202601	1.0	0.875000	NaN	NaN	NaN	NaN
202602	1.0	NaN	NaN	NaN	NaN	NaN

8. divide each row



9. create visualization

IMPACT

Key Findings & Business Implications

The Insight

Customer acquisition drops sharply during the university holiday period, falling from 53 new users in November down to 29 in December, hitting an all-time low of 8 in January and 7 in February. Furthermore, cohorts acquired right before the break (e.g., October and November) suffer their lowest retention as campus activity pauses.

The Business Implication

Operating in the immediate vicinity of a major university campus, this laundry business relies heavily on the dynamics of student activity, catering primarily to college students in nearby boarding houses and residential areas, which leaves its revenue naturally fluctuating between academic terms and holiday breaks.

Thank You!

Note:
All data has been modified and anonymized to protect business confidentiality, while the analysis process remains representative of the actual project.

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